GEOMETRY · UNIT 8: TRANSFORMATIONS

Animation Studio

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students animate a video game sprite on a coordinate grid. They apply translations, reflections, rotations, and dilations with coordinate rules, compose transformations and test whether order matters, describe symmetry, and storyboard an animation of their own.

Coordinate rules make transformations precise and connect geometry to programming. Composition shows that order matters, and the design task uses sequences of transformations.

Standards Addressed

- G-CO.2 Represent transformations as functions
- G-CO.4 Develop definitions of rotations, reflections, and translations
- G-CO.5 Specify a sequence of transformations
- G-SRT.1 Properties of dilations

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch: show a simple game animation. Part 1 single transformations.
Day 2	Part 2 compositions and symmetry.
Day 3	Part 3 storyboard.

Materials

- Student handout
- Graph paper or a transparency
- Colored pencils
- Optional: Desmos or GeoGebra

How the Three Versions Differ

Level	What changes
CP2	Apply each rule in a table and identify rigid motions.
CP1	Compositions, rotations of 90°, 180°, 270°, reflections over y = x, and finding a rule from an image.
Honors	Rotations about a point that isn't the origin, dilations about any center, finding a line of reflection, and compositions of reflections.

Watch For (Common Misconceptions)

- Rotating clockwise when counterclockwise is meant.
- Reflecting over the x-axis by negating x.
- Thinking dilations preserve length.

Extension Ideas

- Code the animation in Scratch or Desmos.
- Make a flip book.
- Explore glide reflections in footprints.

Answer Key – CP2

Part 1: Single Moves

#	Task	Answer
1	Complete the table. Rotate 90° counterclockwise about the origin with (x, y) → (−y, x).	Translate: (6, −1), (8, −1), (6, 2). Reflect: (−1, 1), (−3, 1), (−1, 4). Rotate: (−1, 1), (−1, 3), (−4, 1). Dilate: (2, 2), (6, 2), (2, 8).
2	Write the rule for a reflection over the y-axis and a dilation by 2 centered at the origin.	Reflection over y-axis: (x, y) → (−x, y). Dilation: (x, y) → (2x, 2y).
3	Which transformations keep the sprite the same size and shape (rigid motions)? Which one does not?	Translations, reflections, and rotations are rigid. The dilation changes the size.

Part 2: Combine Moves

#	Task	Answer
4	Reflect the sprite over the x-axis, then translate it up 3. Give the final coordinates.	(1, −1) → (1, 2); (3, −1) → (3, 2); (1, −4) → (1, −1)
5	Now translate up 3 first, then reflect over the x-axis. Are the results the same? What does this tell you?	Translate first: (1, 4), (3, 4), (1, 7); then reflect: (1, −4), (3, −4), (1, −7). Different, so order matters.
6	A logo shaped like a regular hexagon looks the same after a rotation. What is the smallest rotation that works?	60°

Part 3: Storyboard

#	Task	Answer
7	Choose three different moves for frames 2, 3, and 4. Write each rule.	Answers vary.
8	Give the coordinates of the sprite in every frame.	Answers vary.
9	Draw the storyboard on graph paper.	Answers vary.

Answer Key – CP1

Part 1: Single Moves

#	Task	Answer
1	Apply rotations of 90°, 180°, and 270° counterclockwise about the origin to the sprite. Write each rule.	90°: (−y, x) → (−1, 1), (−1, 3), (−4, 1). 180°: (−x, −y) → (−1, −1), (−3, −1), (−1, −4). 270°: (y, −x) → (1, −1), (1, −3), (4, −1).
2	Reflect the sprite over the line y = x. Write the rule.	(x, y) → (y, x): (1, 1), (1, 3), (4, 1)
3	Dilate by ½ centered at the origin. Compare the side lengths and the area with the original.	(0.5, 0.5), (1.5, 0.5), (0.5, 2). Sides are half as long; the area is ¼ as large.

Part 2: Combine Moves

#	Task	Answer
4	Reflect over the x-axis, then over the y-axis. Write one rule for the result. Which single transformation is it?	(x, y) → (−x, −y): a 180° rotation about the origin.
5	Show that reflecting then translating gives a different result from translating then reflecting (use a reflection over the x-axis and a...	Reflect then translate: (1, 2), (3, 2), (1, −1). Translate then reflect: (1, −4), (3, −4), (1, −7).
6	The sprite's image is A′(4, −2), B′(6, −2), C′(4, 1). Describe the transformation and write its rule.	Translation 3 right and 3 down: (x, y) → (x + 3, y − 3)
7	Describe a sequence of two transformations that maps the sprite to A″(−1, −1), B″(−3, −1), C″(−1, −4).	Rotate 180° about the origin (or reflect over the x-axis, then over the y-axis).

Part 3: Storyboard

#	Task	Answer
8	Plan the moves and write each rule.	Answers vary.
9	List the coordinates in each frame and show the last frame equals the first.	Answers vary.
10	Explain why your sequence returns to the start.	Answers vary.

Answer Key – Honors

Part 1: Moves About Any Point

#	Task	Answer
1	Rotate the sprite 90° counterclockwise about A(1, 1).	Subtract A, rotate, add A: B(3, 1) → (1, 3); C(1, 4) → (−2, 1); A stays (1, 1).
2	Dilate the sprite by 3 with center A(1, 1). Is A′ different from A?	A′ = A(1, 1); B′(7, 1); C′(1, 10). The center does not move.
3	Write a general rule for rotating 90° counterclockwise about (h, k).	(x, y) → (h − (y − k), k + (x − h))

Part 2: Reflections & Compositions

#	Task	Answer
4	A reflection maps A(1, 1) to (5, 3). Find the equation of the line of reflection.	Midpoint (3, 2); slope of AA′ is ½, so the line has slope −2: y = −2x + 8.
5	Reflect the sprite over x = 2 and then over x = 5. Show that the result is a translation and find its distance.	x → 4 − x → 10 − (4 − x) = x + 6: a translation 6 units right (twice the distance between the lines).
6	Reflect over the x-axis and then over y = x. Show that the result is a rotation and find its angle.	(x, y) → (x, −y) → (−y, x): a 90° counterclockwise rotation about the origin.

Part 3: Storyboard with Proof

#	Task	Answer
7	Plan the frames and rules.	Answers vary.
8	List all coordinates.	Answers vary.
9	Prove that your loop returns to the start using compositions.	Answers vary.

Scoring Rubric (all levels)

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Animation uses at least three different transformations with correct rules and coordinates. Goes beyond the minimum with a creative, realistic choice.	Animation uses at least three different transformations with correct rules and coordinates.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Storyboard shows each frame on a grid with the coordinate rule and the image coordinates. Polished, organized, and visually engaging.	Storyboard shows each frame on a grid with the coordinate rule and the image coordinates. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.